

An evolutionarily conserved scheme for reformatting odor concentration in early olfactory circuits

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eLife Assessment

This is a **valuable** computational study of odor responses in the early olfactory system of insects and vertebrates. The study addresses the question of how information about odor concentration is encoded by second-order neurons in the invertebrate and vertebrate olfactory system; it offers insights into the transformation of neural signals from receptors to second-order neurons. While reanalysis of published data presents **solid** evidence supporting compression of concentration information, **incomplete** analysis is provided to resolve how this observation could be reconciled with the need to preserve information about changes in stimulus intensity. This work will be of interest to neuroscientists studying sensory processing broadly and olfaction specifically.

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Abstract

Understanding how stimuli from the sensory periphery are progressively reformatted to yield useful representations is a fundamental challenge in neuroscience. In olfaction, assessing odor concentration is key for many behaviors, such as tracking and navigation. Initially, as odor concentration increases, the average response of first-order sensory neurons also increases. However, the average response of second-order neurons remains flat with increasing concentration – a transformation that is believed to help with concentration-invariant odor identification, but that seemingly discards concentration information before it could be sent to higher brain regions. By combining neural data analyses from diverse species with computational modeling, we propose strategies by which second-order neurons preserve concentration information, despite flat mean responses at the population level. We find that individual second-order neurons have diverse concentration response curves that are unique to each odorant — some neurons respond more with higher concentration and others respond less — and together this diversity generates distinct combinatorial representations for different concentrations. We show that this encoding scheme can be recapitulated using a circuit computation, called divisive normalization, and we derive sufficient conditions for this diversity to emerge. We then discuss two mechanisms (spike rate vs. timing based) by which higher order brain regions may decode odor concentration from

the reformatted representations. Since vertebrate and invertebrate olfactory systems likely evolved independently, our findings suggest that evolution converged on similar algorithmic solutions despite stark differences in neural circuit architectures. Finally, in land vertebrates a parallel olfactory pathway has evolved whose second-order neurons do not exhibit such diverse response curves; rather neurons in this pathway represent concentration information in a more monotonic fashion on average, potentially allowing for easier odor localization and identification at the expense of increased energy use.

Introduction

A central task of sensory systems is to reformat representations of sensory stimuli to efficiently encode key stimulus features. For example, in vision, brightness information captured by the retina needs to be preserved along the visual pathway to facilitate accurate perception of light intensity and contrast in the environment (1 [↗](#)). Similarly, in audition, sound intensity captured by hair cells in the cochlea needs to be preserved to facilitate behaviors that require sound localization (2 [↗](#)). While these transformations are well-studied for visual and auditory stimuli (3 [↗](#)–7 [↗](#)), much less is known about analogous transformations in the olfactory system. The ability to reliably estimate odor concentration (i.e., how much of an odor is present) is critical for olfactory navigation across species (8 [↗](#), 9 [↗](#)). Here, we investigate the transformation of this key feature by early olfactory circuits.

Reliably tracking odor concentration is challenging (10 [↗](#)). Based on receptor-ligand interactions, first-order (sensory) neurons increase their firing rates as odor concentration increases (11 [↗](#)–15 [↗](#)), such that the population activity of sensory neurons scales with odor concentration. However, this comes at a cost of odor discrimination, as representations become increasingly overlapping at high concentrations (16 [↗](#)–19 [↗](#)). In contrast, the average population activity of second-order and third-order neurons remains relatively flat over concentration changes that vary over many orders of magnitude (20 [↗](#)–22 [↗](#)). This property is thought to be critical for concentration-invariant identification of an odorant (20 [↗](#), 23 [↗](#)–25 [↗](#)). However, this creates a conundrum: if concentration information is discarded in second-order neurons, which exclusively transmit odor information to the rest of the brain, how does the brain support olfactory behaviors, such as tracking and navigation?

By combining neural data analyses from diverse species with computational modeling, we provide a resolution to this issue and offer four contributions: First, we show how concentration information can be represented in second-order neurons, despite mean response flatness at the population level, and we show that this property is conserved across three species (locusts, zebrafish, and mice). Second, we demonstrate that a unitary circuit computation — divisive normalization — can generate all primary features of the reformatted second-order neuron representation given sufficient conditions on first-order responses. Third, we describe how this reformatted representation can be used to decode odor concentration using two biologically plausible strategies (rate-based and time-based decoding), and we discuss decoding constraints imposed by each scheme. Fourth, in land vertebrates, evolution has invented a parallel olfactory pathway whose response properties are much closer to those of first-order neurons; i.e., responses of second-order neurons in this pathway scale with concentration. We show that this representation substantially improves concentration decoding and odor localization (26 [↗](#)), albeit with higher energy costs. Moreover, we hypothesize that the emergence of a parallel pathway in land vertebrates, in conjunction with the presence of long-range cortical feed-back loops, may have afforded neurons of the first pathway to carry out higher-order computations extending beyond simply representing sensory features.

Together, this study provides a unified theoretical framework for understanding how concentration information is reformatted by early olfactory circuits separated by several hundred-million years of evolution, from insects to mammals. Given the likely independent evolution of olfactory systems in vertebrates and invertebrates, our results suggest that evolution has converged to a common algorithm despite differences in the underlying circuit implementations.

Results

Olfaction begins with olfactory sensory neurons (OSNs) that each expresses a single type of odorant receptor protein that binds with specific odor molecules (**Fig. 1A**). These first-order neurons transmit odor information to second-order neurons, such as projection neurons in insects, or mitral cells in vertebrates. The activity of second-order neurons is a combination of first-order neuron activity, as well as interneuron modifications (e.g., from local interactions in the antennal lobe of insects (27), or from local interactions in the olfactory bulb and long-range feedback in vertebrates (28–30)). Only second-order neurons project to the rest of the brain (e.g., the mushroom body in insects, the olfactory cortex in vertebrates), which subsequently orchestrate complex olfactory behaviors, such as odor discrimination and tracking (31). Hence, it is critical for second-order neurons to preserve key features of odors, including their identity and concentration. To determine if there is a conserved strategy of encoding odor concentration in olfactory circuits, we studied odor responses of first- and second-order neurons across several species. For first-order neurons, we re-analyzed data from fruit flies (24 odorant sensory neurons to 9 fruit odors across 4 concentrations (32), spanning 6 orders of magnitude in relative dilution) and mice (47 glomeruli to 5 odors across 4 concentrations (33), spanning 3 orders of magnitude). For second-order neurons, we re-analyzed data from locusts (110 projection neurons to 3 odors at 5 concentrations (21), spanning 3 orders of magnitude), mice (392 mitral cells to 5 odors at 4 concentrations (34), spanning 3 orders of magnitude) and zebrafish (358 mitral cells to 1 odor at 5 concentrations (22), spanning 4 orders of magnitude).

Shared properties of odor concentration encoding in second-order neurons

We observed two common properties of odor representations in second-order neurons across all three species (**Fig. 1B–D**, top panels). First, for each odor, the average population response of second-order neurons remains nearly flat across concentrations that range 3–4 orders of magnitude. The fitted slopes of average responses of second-order neurons across increasing concentration levels are significantly lower than those of first-order neurons (**Fig. 1E**), consistent with previous results (21, 22, 33–36).

Second, the response curves of individual second-order neurons across increasing concentration levels have diverse shapes (**Fig. 1B–D**, bottom panels). We quantified the diversity of each neuron's concentration response curve by computing the slopes between responses across concentration levels. The change in response across consecutive concentration levels may not be robust due to experimental noise and the somewhat limited range of concentrations sampled. Thus, for better illustration and ease of analysis, we computed the slopes between responses separated by 2 dilution levels, corresponding roughly to a 100x range in concentration. For example, the two slopes per neuron were computed using responses from concentration level 3 to level 1 and from level 4 to level 2. We defined the neuron to have a monotonically increasing (or decreasing) concentration response shape if both slopes are positive (negative). Non-monotonic concentration response shapes represent responses that increase and then decrease, or that decrease and then increase. We then calculated the angle between these two slopes for each neuron and plotted a polar histogram of these angles over all neurons (**Fig. S1**). Each quadrant of the polar histogram represents one of the four possible shapes (top right: monotonically

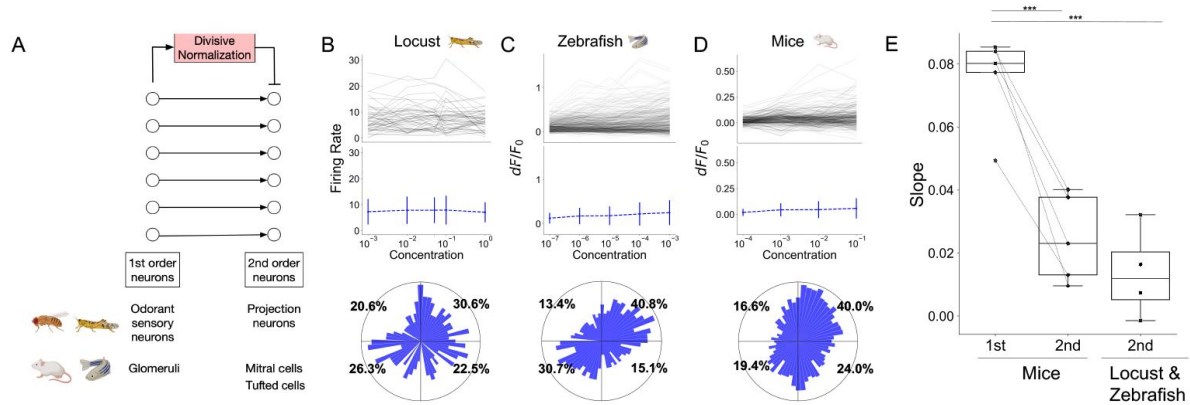


Fig. 1.

Second-order neurons across species demonstrate two common properties: flat average response of the population and diverse response shapes of individual neurons.

(A) Illustration of the early olfactory circuit across species. **(B–D)** Example responses of second-order neurons in locust (to odor geraniol), zebrafish (phenylalanine), and mice (valeraldehyde), respectively. Top: individual neuron response curves across concentrations. Middle: mean population responses across concentrations. Bottom: polar histogram showing the distribution of individual neuron response shapes for all odors combined. The x-axis shows the slope between neuron response at concentration level 1 (lowest) to concentration level 3, and the y-axis shows the slope from concentration level 2 to concentration level 4 (highest). The angle of the bar shows the direction of the vector formed by these two slopes, and the length of the bar shows the percentage of neurons with the same angle. Top right quadrant: monotonically increasing responses; top left quadrant: decreasing then increasing; bottom left quadrant: monotonically decreasing; bottom right quadrant: increasing then decreasing. **(E)** Fitted slopes for mean population responses across concentrations for first- and second-order neurons. Each dot represents a single odor. First two columns show data collected in mice, and dashed lines connect the same odor. The third column shows all odors tested in locust and zebrafish combined. T-test is performed between the slopes formed by first- and second-order neurons; *** p-value < 0.001.

increasing, top left: decrease then increase, bottom left: monotonically decreasing, bottom right: increase then decrease), and the percentage shown in each quadrant shows the percentage of neurons with the corresponding shape. We found diverse response shapes for second-order neurons in all three species (**Fig. 1B–D**). For example, for projection neurons in locust, 30.6% of neurons monotonically increased (top right quadrant), 26.3% monotonically decreased, 20.6% decreased then increased, and 22.5% increased then decreased. The response shapes were similarly spread over the four quadrants for mitral cells in both mice (40.0%, 19.4%, 16.6%, 24.0%) and zebrafish (40.8%, 30.7%, 13.4%, 15.1%). Thus, even though the average responses of second-order neurons are flat across concentrations, the response shapes of individual neurons are diverse.

In summary, two properties of second-order neuron responses across concentrations — flat average response of the population and diverse response shapes of individual neurons — are shared across locusts, zebrafish and mice.

Divisive normalization recapitulates two ubiquitous properties of second-order olfactory neurons

How might such diverse shapes be generated from first-order neurons that respond more monotonically (37, 38)? One clue is the mean flatness of second-order responses, which hints at some type of normalization mechanism. We explored three commonly described mechanisms (22, 36, 39, 40) — divisive normalization (DN), subtractive normalization (SN), and intraglomerular gain control (IGC). Both divisive and subtractive normalization are global schemes based on population level activity, whereas intraglomerular gain control is a local normalization scheme where the activity of a neuron is normalized by only its own activity. We tested whether these normalization mechanisms could generate the two properties of second-order neurons described above (flat average response of the population and diverse response of individual neurons, across concentration levels) starting from experimentally observed first-order neuron responses in fruit flies and mice (**Fig 2Ai–Aii**).

Divisive normalization divides the activity of a single neuron by the summed population activity (22, 36):

$$d_i = r_{\max} \left(\frac{r_i^n}{\sigma^n + r_i^n + k(\sum_j r_j)^n} \right), \quad (1)$$

where d_i is the activity of the i^{th} second-order neuron, r_i is the activity of the corresponding i^{th} first-order neuron, $r_{\max}$ is the estimated maximum response of first-order neurons (determined individually for each dataset), $\sigma > 0$ controls the neuron's sensitivity to normalization, $k \in (0, 1)$ controls the strength of normalization, and $n > 0$ controls the shape of normalization curve. See Methods for parameter settings and equations for SN and IGC.

We found that after divisive normalization was applied to experimental first-order neuron responses, the simulated second-order neuron responses became significantly flatter over concentrations in both flies (1st order neuron slopes: 0.05 ± 0.01 vs. 2nd order slopes: 0.00023 ± 10^{-4} ; $p < 0.001$, **Fig. 2Bi, Ei**) and in mice (1st order: 0.08 ± 0.01 vs. 2nd order: 0.02 ± 0.01 ; $p < 0.01$, **Fig. 2Bii, Eii**). Thus, the average population responses of the simulated second-order neurons are nearly flat across concentrations, generating the first property. For the second property, we found that simulated second-order response shapes become more uniformly distributed in the polar plot histogram compared to those of first-order neurons. For example, the percentage of monotonically increasing neurons dropped from 58.3% in first-order neurons to 29.2% in second-order neurons in flies (**Fig. 2Bi**) and from 70.0% to 50.0% in mice (**Fig. 2Bii**). Thus, divisive normalization can reproduce both properties of second-order neurons.

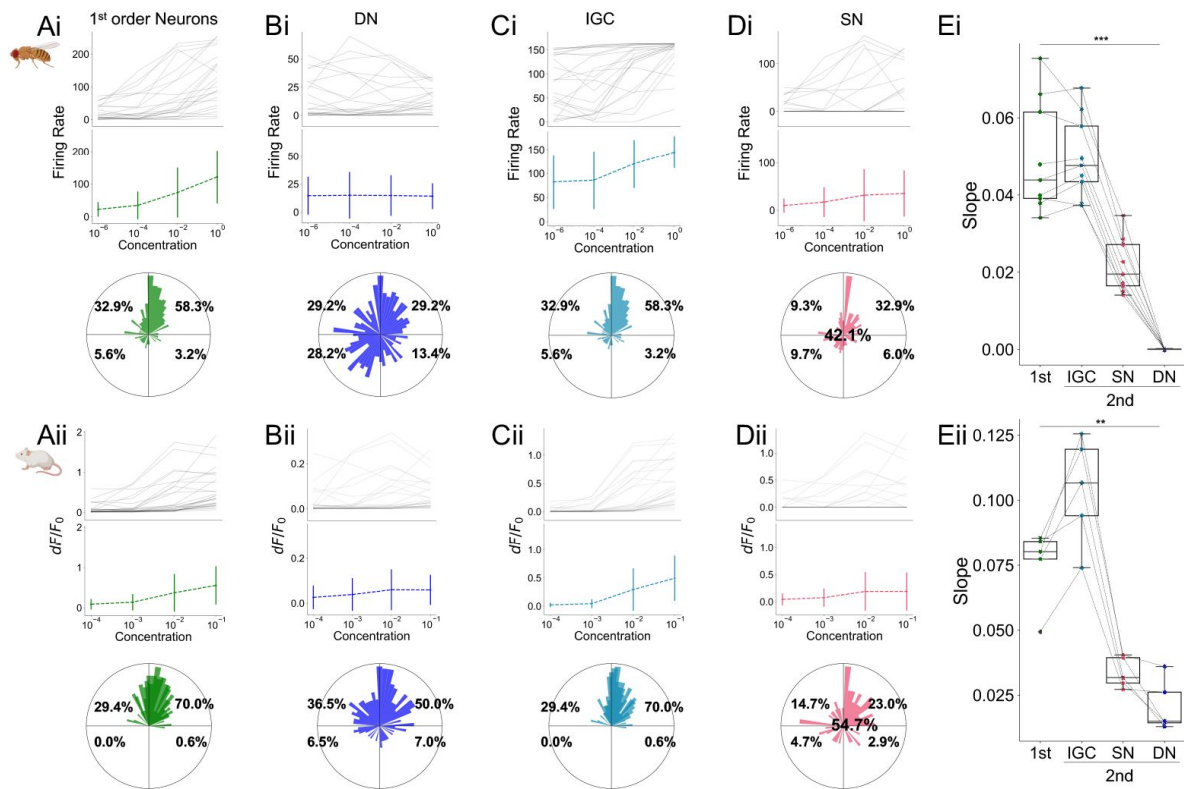


Fig. 2.

Responses of simulated second-order neurons recapitulate the two common properties.

(Ai-Aii) Example experimentally measured responses of first-order neurons in fruit flies (apple odor) and mice (heptanal). **(Bi-Bii)** Simulated responses of corresponding second-order neurons after applying divisive normalization (DN) to first-order neurons. The top two rows show results of a single example odor, while the bottom row (the polar plots) show all odors combined. **(Ci-Cii)** Same as B but after applying subtractive normalization (SN). **(Di-Dii)** Same as B after applying intraglomerular gain control (IGC). **(Ei-Eii)** Fitted slopes for the mean population response across concentrations for first-order neurons, as well as simulated second-order neurons after IGC, SN, or DN is applied. Each dot represents an odor. Dashed lines connect the same odor across normalization methods. Paired t-test is performed between the slopes formed by first- and second-order neurons; *** p-value < 0.001, ** p-value < 0.01.

In contrast, neither subtractive normalization nor intraglomerular gain control reproduce both properties of second-order neurons (**Fig. 2C–E**). Subtractive normalization recapitulates mean flatness at the population level in a relatively small parameter window (at the cost of silencing a large portion of neurons); however, it cannot reliably reproduce diverse response curves. Intraglomerular gain control fails to generate either property.

While divisive normalization has been previously shown to generate flat mean responses at the population level (22, 36), its role towards generating diverse concentration response curves has not been appreciated, and this latter property, as we will show, is critical towards decoding concentration levels downstream.

Sufficient conditions for generating diverse concentration response curves via divisive normalization

Do these two properties inevitably arise as a result of divisive normalization applied to *any* first-order neuron responses, or do first-order responses need to have certain structure them-selves? To answer this question, we synthesized concentration response curves for first-order neurons. These responses (r) to logarithmic scale concentration levels (x) follows the logistic function (37, 38):

$$r_i = R_i \left(\frac{1 - s_i}{1 + \exp(-a_i(x - b_i))} + s_i \right), \quad (2)$$

where R_i is the saturation level of neuron i , a controls the rate of saturation, b determines the location where the response of the neuron reaches half-maximum, and s is a small positive parameter representing the amount of spontaneous activity in the absence of stimuli.

We explored two classes of first-order neuron concentration responses. The first class does not exhibit cross-overs; i.e., the rank order of neurons, from the highest response amplitude to the lowest, remains the same across all concentrations. The second class of concentration responses exhibited cross-overs, which are closer to experimental first-order neuron responses due to combinatorial odor binding properties of OSNs (12–15, 32, 41). Here, the response amplitude ranks change, reflecting different rates of activity increase in neurons despite the overall trend of increasing. To simulate no cross-over responses (**Fig. 3A**), the values of a , b , and s are fixed for all neurons. To simulate cross-over responses (**Fig. 3B**), each simulated neuron has different values of a , b , and s sampled from independent uniform distributions. For both classes, the saturation level (R) is sampled from a Gamma distribution, which fits the experimentally observed activities of first-order neurons (Methods). By construction, for both classes, individual and average first-order neuron responses grow monotonically with increasing concentration level (**Fig. 3A–B**).

We found that the observed diversity of response curves in second-order neurons could only be reproduced after applying divisive normalization to first-order responses that had cross-overs, as opposed to no cross-overs (**Fig. 3C–D**). Specifically, for the former, the bars in the polar plot histogram of second-order neuron responses span all quadrants, whereas for no cross-over responses, the bars are exclusively located in the upper-right quadrant (i.e., monotonically increasing).

Why do diverse response curves emerge only from cross-over first-order responses? Divisive normalization (Eqn. Eq. (1)) effectively calculates the relative contribution of a neuron — i.e., its activity divided by the total activity of all neurons in the population — since the summation term $k(\sum_j r_j)^n$ dominates the $\sigma^n + r_i^n$ term in the denominator. Therefore, divisive normalization mostly does not alter the relative contribution (rank order) of each neuron in the ensemble. To create diverse shapes in second-order neurons, the first-order neurons must themselves increase their

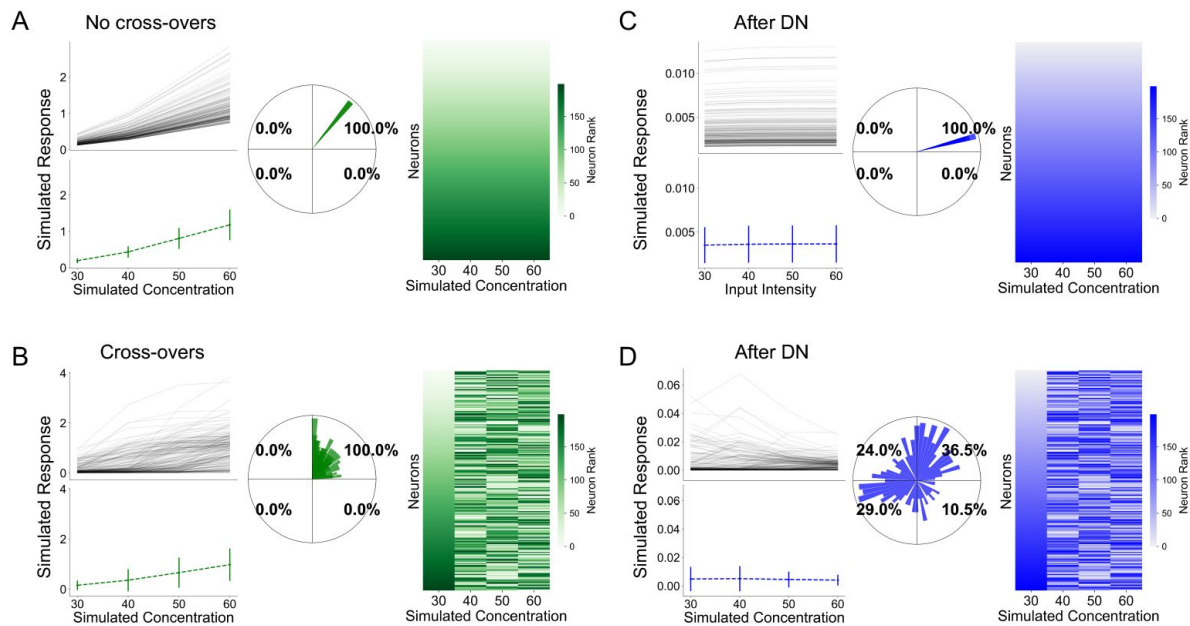


Fig. 3.

Sufficient conditions to generate diverse concentration response shapes using divisive normalization.

(A–B) Simulated response curves of first-order neurons across concentration levels. Two classes of first-order responses are considered: those with cross-overs and those without cross-overs. Left: single neuron response curves and population average across four simulated concentration levels. Center: polar histogram for the shapes of response curves. Right: rank order of neuron activities at each concentration level (neurons are sorted according to response strength at the lowest nominal concentration, and the same indexing is preserved across the higher concentrations). **(C–D)**. Same as A and B but showing the responses of simulated second-order neurons after divisive normalization is applied to first-order neurons.

responsiveness with sufficient diversity (faster or slower, or steeper vs. shallower, compared to the average changes in activity of the population) such that their rank orders change across concentration levels. Biologically, as concentration increases, cross-overs emerge as a consequence of competing growth rates of individual first-order neurons versus the population. In summary, cross-over first-order responses are sufficient for divisive normalization to generate non-monotonic responses in second-order neurons.

Two complementary schemes of decoding odor concentration from populations of second-order neurons

How can higher order brain regions decode concentration information from second-order neurons that have flat mean responses, but diverse individual concentration response curves? We discuss two potential strategies, one using firing rate information and the other using spike timing information.

First, we explored how concentration information can be decoded from the diverse shapes of response curves of second-order neurons. We used a simple multi-class logistic regression model to predict the concentration of an odor using the experimental odorant response activities of second-order neurons, where the model was trained using all but one trial and tested on the held-out trial (Methods). We found that accuracies were nearly-perfect in all three species (**Fig. 4A**): 100±0% in locusts, 96±4% in zebrafish, and 97±4% in mice (chance accuracy is 25% for data with four concentration levels, and 20% for five concentration levels). More-over, this decoding accuracy is on par with that derived using first-order neuron responses (**Fig. 4B**) and corresponding simulated responses of second-order neurons after divisive normalization is applied (**Fig. 4C**). Hence, divisive normalization reformats concentration information, so that it is no longer coded by the amplitude of mean population responses, but by the diverse shapes of response curves. In other words, each concentration level has its own combinatorial code in second-order neurons, allowing for the concentration to be decoded.

Second, we considered a complementary scheme by which odor concentration can be decoded using information in the time-domain. Latency or primacy coding (42–44) proposes a mechanism to achieve concentration-invariant odor recognition by using information in only a few second-order neurons that have the highest affinity to an odor, i.e., neurons that respond early and stably to the odor across concentrations. To determine if this mechanism could in addition decode concentration levels, we simulated the temporal responses of 200 first-order neurons to three concentration levels of an odor using logistic functions with different parameter values; e.g., one parameter value controls the sensitivity of the response curve (**Fig. 4D**; Methods). After divisive normalization, the logistic response curves of first-order neurons transformed into non-monotonic curves (**Fig. 4E**). This agrees with the temporal response shapes of second-order neurons observed experimentally (45, 46).

We report three observations from these simulations. First, the set of second-order neurons with the highest-affinities (where affinity is based on the time latency of response) remained almost the same across concentrations levels: the Jaccard similarity between the sets of highest-affinity neurons for each pair of concentration levels was > 0.96, **Fig. 4F**). Second, the average timing of the peak response (i.e., the response latency) of these primacy neurons, decreased as concentration increases (**Fig. 4G**). This is not surprising and is simply meant to model the binding of odorants to odorant receptors with its resulting inverse correlation between response amplitude and latency (higher amplitude shorter latency) (47). Third, critically, the rank orders of the set of primacy neurons were scrambled across concentrations, due to cross-over responses of corresponding first-order neurons. For the 10 and 50 highest-affinity neurons responding to the high concentration level, 7 (70%) and 32 (64%) shifted ranks, respectively, compared to their corresponding ranks at low concentration; for all 200 simulated neurons, 164 (82%) shifted ranks from high to low concentration (**Fig. 4H**).

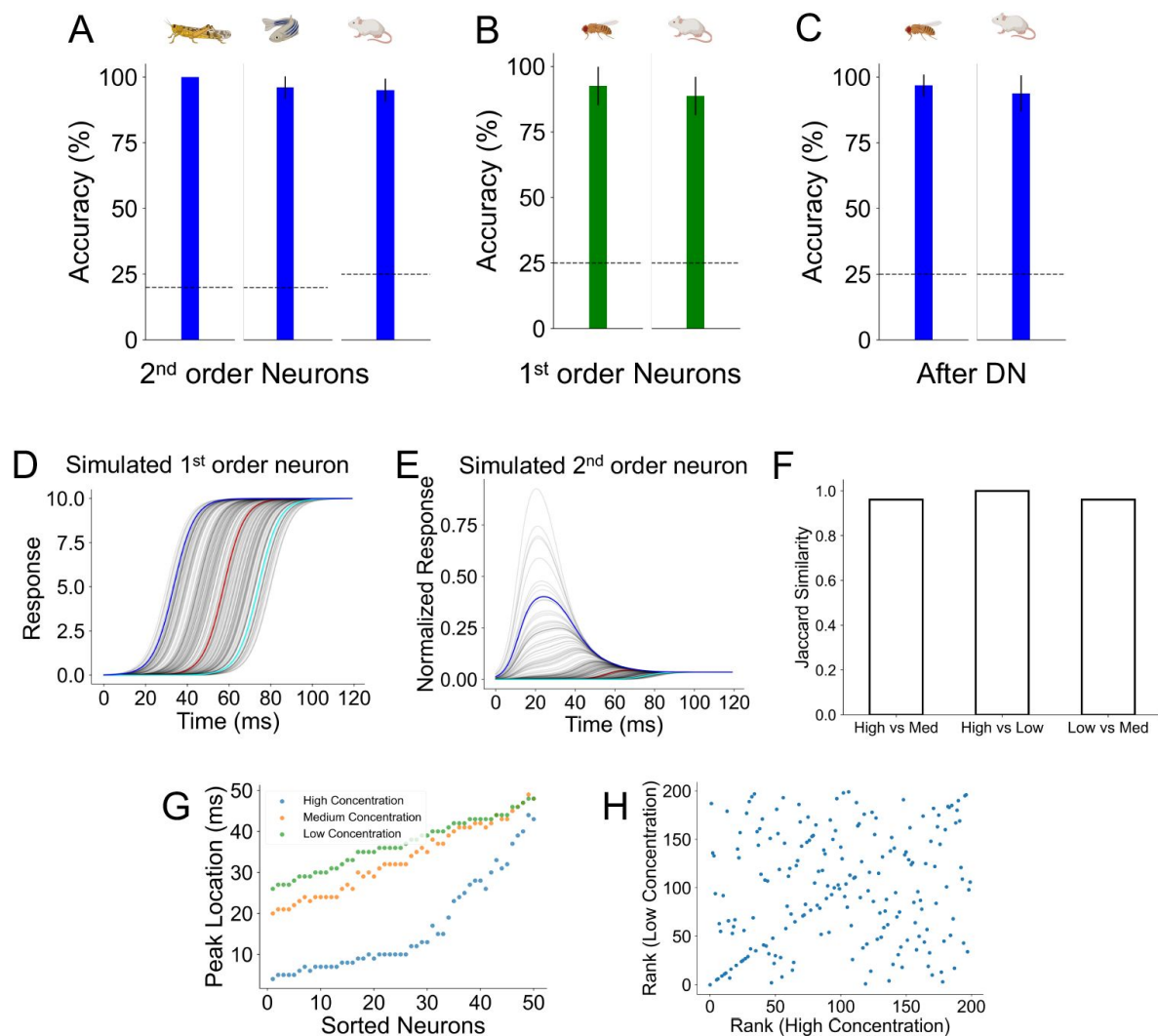


Fig. 4.

Divisive normalization reformats concentration representations in the rate and time domains.

(A) Accuracy for concentration classification using experimental responses of second-order neurons (left: locust; center: zebrafish; right: mice). The concentration levels of each odor are classified independently per odor. Reported are the average accuracy across all odors, and error bars show standard deviations. Dashed line shows the accuracy level for random guess (0.2 for locust and zebrafish, 0.25 for mice). (B) Same as A but for experimental first-order neurons in fruit flies and mice. (C) Same as A but after divisive normalization applied to first-order responses in B. (D) The temporal response curves of simulated first-order neurons to an odor. Neurons with different affinity to this synthetic odor are highlighted (high affinity: blue; medium affinity: red; low affinity: cyan). (E) Same as D, after divisive normalization is applied. (F) Jaccard similarity between the sets of the 50 earliest-responding neurons for each pair of concentration levels. The value of Jaccard similarity ranges between 0 and 1; 1 means the two sets of neurons contain identical neurons, and 0 means they share no neurons in common. (G) The time when the 50 earliest-responding neurons respond under each concentration level. The neurons are sorted by their peak time under low concentration. (H) Rank of neurons under low concentration based on their time of peak response (y-axis) vs. rank of neurons under high concentration (x-axis).

These observations have major implications for both concentration-invariant odor identification and odor concentration decoding by third-order neurons under the primacy model. Odor identity can be encoded by the set of highest-affinity neurons (which remains invariant across concentrations); however, the downstream read-out mechanism must not be sensitive to their rank order. Odor concentration can be encoded using either the average response latency of neurons (which scales with concentration), or alternatively, by learning to associate different temporal sequences of neuronal responses with different concentrations.

Hence, two complementary strategies emerge in decoding odor concentration by higher-order brain regions. Divisive normalization, operating in antennal lobe (insects) or olfactory bulb (mammals) circuits, is a potential mechanism for generating odor-specific subsets of second-order neurons whose combinatorial activity or whose response latencies represents concentration information. Importantly, in both schemes, some form of learning within downstream neural circuits is required.

A parallel pathway for odor representation in land vertebrates better supports odor localization

Locating an odor source is essential for survival in the wild. While olfactory navigation over large distances can be solved by serial sniffing ([48](#)), at close distances rodents rely more on stereo olfaction to localize stimuli ([49–52](#)). This requires that the brain quickly determine: 1) if there exists a difference in odor concentrations between two sniffs or between signals from two nostrils in a single sniff; 2) if there is a difference, which sniff/nostril sensed a higher concentration; and 3) how strong the difference is.

We showed above that odor concentration could be decoded from second-order neurons, but the proposed mechanisms require additional learning in downstream neural circuits, which may be slow. Odor localization may be achieved more easily and quickly (e.g., by a simple comparator across nostrils) if a direct copy of concentration information, where neuronal responses increase monotonically with concentration, could be stored beyond first-order neurons.

Indeed, evolution has discovered a parallel pathway in land vertebrates involving an additional type of second-order neuron — the tufted cells. Compared to mitral cells, tufted cells on average respond faster and more reliably to lower concentrations of an odorant, and their activity increases nearly monotonically with increasing odor concentration (19, 26, 53–57) (**Fig. 5A–B**), resembling first-order neuron (glomerular) activity. Previous work showed that tufted cell ensembles outperform mitral cell ensembles in concentration classification and odor generalization tasks ([26](#)).

Here we show that in addition to these tasks, tufted cells are better suited to solve the stereo odor localization problem (**Fig. 5C**, left). We re-analyzed second-order neurons in mice ([34](#)), consisting of 392 mitral cell and 387 tufted cell responses to 5 odors across 4 concentration levels that span 3 orders of magnitude in relative dilution. We found that the same concentration change causes a significantly larger change in tufted cell responses compared to mitral cells (**Fig. 5D**). This predicts that tufted cells should better discriminate smaller concentration changes. Relying on experimentally measured fall-off of concentration with distance from an odor source (**Fig. 5C**, right), we indeed find that the tufted cells are better at resolving smaller distances — the same increase in the distance from the odor center would lead to larger change in tufted cell activity compared to mitral cells (**Fig. 5E–F**). However, this increased performance comes at a cost; the average population activity level of tufted cells is roughly three times higher than in mitral cells (0.03 ± 0.01 for mitral cells vs 0.09 ± 0.03 for tufted cells), indicating that tufted cells consume more energy than mitral cells. Thus, we posit that the evolution of the tufted cell output channel traded-off energy for faster and finer concentration discrimination, which is essential for odor tracking and navigation in land vertebrates.

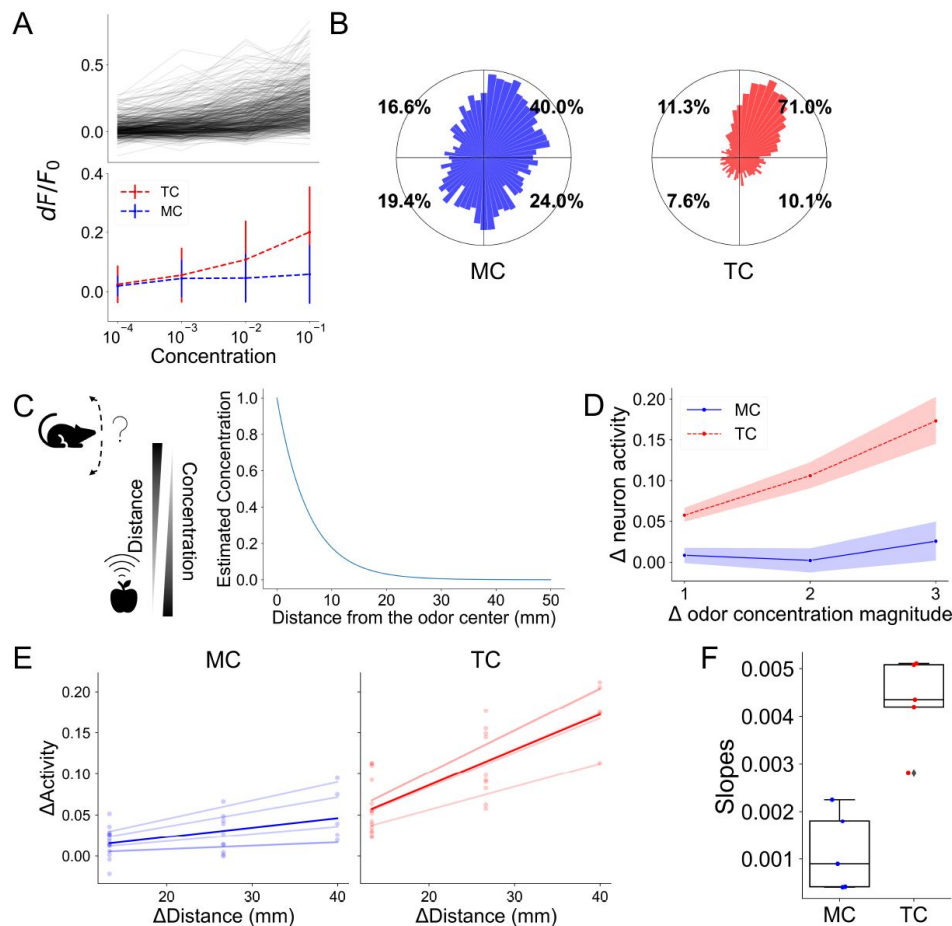


Fig. 5.

A parallel olfactory pathway in land vertebrates provides improved concentration decoding.

(A) Top: concentration response curves of individual tufted cells to a single odor (valeraldehyde). Bottom: average population responses for mitral (MC, blue) and tufted (TC, red) cells. **(B)** Polar histogram showing the distribution of concentration response shapes MCs and TCs for all five odors combined. **(C)** Left: Cartoon illustration for the odor localization task. Right: estimated decrease in odor concentration as distance from the odor source increases. **(D)** Change in average population neuron activities across all pairs of concentrations. The x-axis shows the absolute value of difference in the relative dilution (in order of magnitude) between the two concentration levels. The y-axis shows the change in activity. Results are averaged over five odors, and the shaded area shows the 95% confidence intervals. **(E)** Estimated change of neural activities as the distance between the odor source and the animal's location increases for mitral cells (left) and tufted cells (right). Each dot shows a particular concentration for an odor. There are 20 dots in total (4 concentrations x 5 odors). Each opaque line is a linear regression calculated based on all concentrations of a given odor, and the darker lines are the average over all odors. **(F)** Slopes for the linear regression lines shown in E. Each dot represents an odor. The value of the slope is an estimate of neural sensitivity to the change in distance to the odor source. The higher the slope, the higher the sensitivity.

Discussion

Summary of findings

Differentiating odor concentration is critical for olfactory behaviors, including odor-based navigation and localization. First-order olfactory neurons exhibit increasing responses to higher levels of concentration, but second-order neurons on average remain flat across many orders of magnitude of concentration fluctuations. Such concentration invariance is thought to be beneficial for identifying odorants independent of concentration (20 [↗](#), 25 [↗](#)). However, under this model, it has remained unclear how odor concentration information can be readily accessed by higher brain regions to guide behavior and decision-making. Our analysis, based on previous experimental work (22 [↗](#), 33 [↗](#), 35 [↗](#), 36 [↗](#)), shows that circuits in the early olfactory system across species (locusts, zebrafish, and mice) normalize overall activity of second-order neurons, while also retaining adequate diversity in the individual second-order neuron concentration response curves: some neurons increase or decrease their activity monotonically with concentration, whereas others respond non-monotonically. This diversity enables individual concentration levels to be encoded combinatorially, which can then be decoded using spike rates or spike times. Moreover, we analyzed an additional type of second-order neuron (tufted cells) that has evolved in land vertebrates and that outperforms mitral cells in concentration encoding (26 [↗](#)) and odor localization tasks. Indeed, recent work suggests that the anterior olfactory nucleus, which gets stronger input from tufted cells than mitral cells, plays a central role in decoding odor concentration (26 [↗](#), 58 [↗](#)). These results suggest a trade-off between concentration decoding and normalization processes, which prevent saturation and reduce energy consumption.

What might be the benefits of evolving a parallel second-order output channel (tufted cells) in land vertebrates? We speculate that this alternative pathway may have enabled mitral cells to evolve new functionalities. By combining sensory information from first-order neurons with feedback from higher brain regions, mitral cells may be ideally suited to perform learning-dependent computations beyond simply relaying sensory input, such as predictive processing or integrating context within decision-making (59 [↗](#)). It is also possible that the brain uses multiple concentration decoding strategies in parallel, depending on the complexity of the olfactory scene and the behavioral needs of the moment. Overall, our findings highlight the importance of an evolutionarily conserved computation in odor coding and how the key feature of odors is efficiently preserved along the olfactory pathway.

Mechanisms for concentration decoding

Both concentration decoding schemes proposed here (spike-based and time-based) require learning in downstream circuitry — i.e., third-order neurons need to learn which temporal sequences of neurons are to be associated with each concentration level of an odor. This sequence may be further altered by noise present in natural environments, which may lead to additional variability across repeated observations of the same odorant concentration. Experimental results indeed show that while some neurons remain rank-stable, different subsets of early responding secondary neurons are silenced or start firing at higher vs. lower concentrations of the same odor (19 [↗](#), 26 [↗](#), 42 [↗](#), 60 [↗](#)). As such, a substantial fraction of second-order neurons change rank depending on the concentration range and odor identity. In addition, the spiking temporal integration window of third-order neurons in downstream brain regions ought to be small enough to read-out the temporal shifts in response latencies across concentrations for primacy neurons. The complementary combinatorial response (firing rate) scheme also necessitates learning: different third-order neurons need to be associated, presumably via synaptic plasticity, with different concentrations of a given odorant. The feasibility of the learning algorithms for both decoding schemes remain to be tested experimentally.

Our work does not rule out alternative mechanisms for encoding concentration in olfactory circuits. For example, concentration can be encoded via the total activity in the trigeminal nerve, through the different latencies between the medial and lateral olfactory bulb under different concentrations (61 [↗](#)), or via temporal integration via serial sniffs (counting the total number of odor packets over multiple sniffs) (48 [↗](#), 62 [↗](#)).

Experimental predictions

Our work raises two testable experimental predictions about the effects of disrupting circuitry involved in implementing divisive normalization on downstream encoding and behavior. First, the properties of second-order neurons, such as mean-flatness and diverse concentration response shapes enabled by divisive normalization, would disappear if the neural circuitry supporting divisive normalization is suppressed. In insects, experimental evidence suggests that divisive normalization is likely achieved through the network of inhibitory interneurons in the antennal lobe (63 [↗](#)). In vertebrates, divisive normalization is likely implemented by short-axon cells (GABAergic-dopaminergic inhibitory cells in the olfactory bulb), whose axons provide interglomerular crosstalk (22, 33, 64–68). Indeed, three properties of short-axon cells in the olfactory bulb make them ideally suited to implement divisive normalization: odor responses of short-axon cells scale monotonically with concentration, they functionally inhibit mitral cells, and crucially, ablating these neurons shift the concentration response curves of mitral cells from being diverse to becoming more monotonic (33 [↗](#)). Concomitantly, this loss of diverse response shapes in second-order neurons may hinder fine concentration discrimination of the same odor. Second, for mammals, prior work has hypothesized that concentration-invariant encoding requires recurrent processing in the piriform cortex (20 [↗](#), 44 [↗](#)). In contrast, our results predict that concentration invariant neural responses, the diversity in individual neuron concentration responses, and the proposed latency/primacy representations for odor identification emerge due to computations within the olfactory bulb. Consequently, we predict that these properties will persist even after inactivating the cortex.

Evolutionary conserved transformation of concentration encoding

Is there a shared computation employed by early olfactory circuits from flies to mammals to generate our proposed concentration encoding scheme? Our results indicate that divisive normalization is sufficient to explain two experimentally observed properties of second-order neurons: flat mean responses and diverse individual concentration response curves. As noted above, experimental evidence suggests that divisive normalization is implemented by distinct inhibitory circuits in the antennal lobe in insects vs. the olfactory bulb in mammals. Thus, while the circuit implementation of this transformation may be different across species that diverged over 100 million years ago, invoking Marr, the algorithmic description appears to be conserved. Given that the invertebrate olfactory receptors have evolved independently from those of vertebrates, our results also imply a functional convergence shaped by similar computational goals. Finally, in machine learning, applying divisive normalization to early encoding layers generates efficient coding representations (69 [↗](#), 70 [↗](#)) that accelerate network training (71 [↗](#)), revealing the broad applicability of this transformation in both biology and engineering.

Methods

Analysis of experimental data

Mice

The experimental data was published in Chae et al. (26) (2022). Briefly, head-fixed, naïve mice were presented with 5 odors at 4 concentration levels (across three orders of magnitude), and two-photon calcium imaging was performed to measure responses from 392 mitral cells and 387 tufted cells. We also re-analyzed a dataset consisting of neuronal responses from 47 glomeruli to the same odor panel (33). Each measurement was repeated for 4-6 trials for each odor at each concentration. For each measurement, a monomolecular odor was presented for 4s, preceded by 10-12s of air baseline, followed by 7-10s air recovery. Responses (df/f) were calculated based on the air baseline period, which was determined for each neuron independently. Neuronal responses for each cell-odor pair were defined as the average df/f over the entire odor period. Data from all field of views and all odors were combined in our analysis.

Zebrafish

The zebrafish data was published by Zhu et al. (22) (2013). We re-analyzed calcium imaging responses of 358 mitral cells to a single odor across 5 concentration levels. Mean neuron responses were defined as the df/f for the period after odor onset (about 2s).

Locust

The locust data was published by Stopfer et al. (21) (2003). We re-analyzed spiking activity of 110 projection neurons in response to 3 odors across 5 concentration levels per odor. Each experiment was repeated for 15 trials. Each trial began with a 2s air baseline period, followed by odor presentation for 1s.

Fruit fly

The *Drosophila* data was published by Hallem and Carlson (32) (2006). Spiking activity of 24 ORNs to 9 fruity odors across 4 concentrations were measured. Each ORN measurement was made in a different animal. All analyses were based on average responses of single neurons across the 6 trials (as shown in Table S2 of Hallem and Carlson (32)), except for the classification task of concentrations (Fig. 4B-C), in which the 6 trials were used individually.

Selecting responding neurons

For a given odor and given concentration, a significant neuronal response was determined by activity 2 standard deviations higher than the baseline rate for at least 20% of the trials (e.g., at least 3 trials out of total 15 trials, or 100% of the trials if there is only one trial). Additionally, if a neuron responds to at least one concentration for a given odor, the neuron is considered to be responding to the given odor. Only responding neurons are used for analysis.

Normalization methods

Divisive Normalization (DN)

Divisive normalization is performed according to the equation below:

$$d_i = r_{\max} \left(\frac{r_i^n}{\sigma^n + r_i^n + k(\sum_j r_j)^n} \right),$$

where d_i is the activity of the i^{th} second-order neuron, r_i is the activity of the corresponding i^{th} first-order neuron, $r_{\max}$ is the estimated maximum response of first-order neurons (determined individually for each dataset), k is a parameter controlling the strength of normalization, and n is a

parameter controlling the shape of normalization curve. We fix $n = 1.5$ following Olsen et al. (36), and we fix $k = 0.1$, though we find no qualitative differences in our conclusions using different values of k (Fig. S2). Similarly, we use $\sigma = 12$ for the *Drosophila* data following Olsen et al. (36), and $\sigma = 1$ for all other datasets. This model of divisive normalization was modified from Olsen et al. (36).

Intrglomerular Gain Control (IGC)

Intrglomerular transformation modulates the responsiveness of a neuron based only on its own response as:

$$g_i = r_{\max} \frac{r_i^n}{\sigma^n + r_i^n},$$

where r_i is the activity of the i^{th} first-order neuron, and g_i is the normalized activity of the corresponding second-order neuron. All other constants are fixed as above.

Subtractive Normalization (SN)

In subtractive normalization, each second-order neuron receives inhibition from interneurons proportional to the sum of first-order neuron activities:

$$s_i = \max(0, k \sum_j r_j),$$

where k is the parameter controlling the strength of normalization, r_i is the activity of the i^{th} first-order neuron, and s_i is the normalized activity of the corresponding second-order neuron. We fix $k = 1/N$ for subtractive normalization, where N is the number of first-order neurons.

Fitting slopes for mean population responses

The slopes of population mean responses (Fig. 1E, Fig. 2Ei–Eii) are fitted in a log-linear manner using `LinearRegression` function in the `sklearn` library of Python. The slope of each odor was fitted independently. Before fitting, the mean values were first min-max normalized by the minimum and maximum values of individual neurons.

OSN simulation

The 200 OSN curves are simulated using a logistic function:

$$r_i = R_i \left(\frac{1 - s_i}{1 + \exp(-a_i(x - b_i))} + s_i \right). \quad (3)$$

The parameter R_i (one per neuron) is sampled from a gamma distribution $\Gamma(\alpha, \beta)$, where $\alpha = 1.15$, $\beta = 1.92$, parameters fitted from experimental data in mice glomeruli (). The a parameter controls how fast the curve reaches saturation value R (higher values of a indicate faster saturation but slower activity initiation). The b parameter controls the x value where the OSN activity reaches half-maximum of $r_{\max}$. The s parameter is a small non-zero parameter representing the amount of spontaneous activity in the absence of stimuli.

Concentration-response curves

When simulating concentration-response curves, the parameter x in Eqn. 3 is set to discrete concentration levels at (arbitrary) values of 30, 40, 50 and 60.

Non-crossover responses

The values of parameters a , b , and s are fixed at $a = 0.1$, $b = 50$, $s = 0$.

Crossover responses

The values of a , b and s are sampled from uniform distributions between $[0.05, 0.4]$, $[30, 80]$ and $[0, 0.05]$ respectively.

Temporal response curves

When simulating temporal-response curves, the parameter x in Eqn. Eq. (3) indicates the time ranging from 0 to 120 ms. The parameters for different concentration levels are:

Concentration level	r_{max}	a	b	s
Low	5	0.3	Uniform[30,80]	0
Medium	10	0.2	Uniform[30,80]	0
High	15	0.1	Uniform[30,80]	0

Logistic regression on concentration decoding

Multi-class logistic regressions were performed using the LogisticRe-gression function in the sklearn library of Python, with liblinear kernel and f_2 regularization, and under the “one vs. rest” scheme.

For mice data, concentrations were first classified within each odor independently, and the reported accuracy was averaged over the five odors. There are four trials per given odor-concentration pair for glomeruli data, and five trials for mitral cell data; the regression model was trained on three/four trials randomly selected and tested on the left-out trial. This process was repeated ten times (with a different, random trial left out per concentration level each time), and the final accuracy for this odor is the average over the ten repeats. The concentrations in locust and fruit fly data were classified in the same way as mice data.

For zebrafish data, since there is only one measurement for each odor-concentration pair, trials for each concentration level were simulated by adding noise to the experimental data. In total 10 repeats were simulated for each concentration level. Noise was generated from a Gaussian distribution with mean equal to 0 and standard deviation equal to the mean response of all neurons in the given concentration.

Estimating distance based on odor concentration

The estimates of concentration fall-off as a function of distance from the odor source were based on experimental measurements (Albeanu lab, unpublished data). The mean odor intensity was measured using a photoionization detector (Aurora scientific) while laterally displacing the odor source from the PID in increments of 5 mm, over a total range of 50 mm, across 3 different odors. The observed relationship was well fit by the equation:

$$C(x) = \exp(-0.1729x), \quad (4)$$

where $C(x)$ is the normalized concentration at distance x , and x is the distance (in mm) between the location of measurement and the source of odor.

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Additional files

Supplement Information [↗](#)

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Reviewer #1 (Public review):

Summary

This article is about the neural representation of odors in the early olfactory system of insects, fish, and rodents. Specifically, it regards the transformation that occurs between the olfactory sensory cells and the second-order neurons (projection neurons in insects, mitral/tufted cells in vertebrates). The central question is how the nervous system can encode both the identity of an odor and its concentration over many log units. The authors reanalyze data from experimental studies of odor responses in primary and secondary neurons, and test a range of computational models as to whether they match the observed transformation. They focus on two aspects of the second-order neuron response to odor concentration: the average activity across all neurons varies only a little with odor concentration, and different neurons have concentration-response curves with different shapes. They conclude that a model of divisive normalization can account for these effects, whereas two alternative models fail the test. A second observation is that tufted cells in the rodent system seem to

undergo less normalization than mitral cells, and some reasons for this difference are proposed.

Strengths:

- (1) The work compares different models for normalization, rather than simply reporting success with one.
- (2) The analysis is applied to very diverse species, potentially revealing a common principle of olfactory processing.

Weaknesses:

- (1) It is unclear that animals actually have a need to represent odor concentration over many log units in support of olfactory behaviors.
- (2) The stimuli used in the chosen experiments, and the measure of neural response, are only weakly related to any ecological need, e.g., during odor tracking.
- (3) Some of the comparisons between receptors and second-order neurons also compare across evolutionarily distant insect species that may not use the same coding principles.
- (4) The analysis ignores the dynamics of odor responses, which figure prominently in previous answers to the question of identity/intensity coding.
- (5) There is considerable prior consensus in the literature on the importance of normalization from primary to secondary neurons.

Elaboration of my comments:

(1) Motivation

The article starts from the premise that animals need to know the absolute concentration of an odor over many log units, but the need for this isn't obvious. The introduction cites an analogy to vision and audition. These are cases where we know for a fact that the absolute intensity of the stimulus is not relevant. Instead, sensory perception relies on processing small differences in intensity across space or time. And to maintain that sensitivity to small differences, the system discards the stimulus baseline. Humans are notoriously bad at judging the absolute light level. That information gets discarded even before light reaches the retina, namely through contraction of the pupil. Similarly, it seems plausible that a behavior like olfactory tracking relies on sensing small gradients across time (when weaving back and forth across the track) or space (across nostrils). It is important that the system function over many log units of concentration (e.g., far and close to a source) but not that it accurately represents what that current concentration is [see e.g., Wachowiak et al, 2025 Recalibrating Olfactory Neuroscience..].

Still, many experiments in olfactory research have delivered square pulses of odor at concentrations spanning many log units, rather than the sorts of stimuli an animal might encounter during tracking. Even within that framework, though, it doesn't seem mysterious anymore how odor identity and odor concentration are represented differently. For example, Stopfer et al 2003 showed that the population response of locust PNs traces a dynamic trajectory. Trajectories for a given odor form a manifold, within which trajectories for different concentrations are distinct by their excursions on the manifold. To see this, one must recognize that the PN responds to an odor pulse with a time-varying firing rate, that different PNs have different dynamics, and that the dynamics can change with concentration. This is also well recognized in the mammalian systems. Much has been written about the topic of dynamic coding of identity and intensity - see the reviews of Laurent (2002) and Uchida (2014).

(2) Conceptual

Given the above comments on the dynamics of odor responses in first- and second-order neurons, it seems insufficient to capture the response of a neuron with a single number. Even if one somehow had to use a single number, the mean firing rate during the odor pulse may not be the best choice. For example, the rodent mitral cells fire in rhythm with the animal's sniffing cycle, and certain odors will just shift the phase of the rhythm without changing the total number of spikes (see e.g., Fantana et al, 2008). During olfactory search or tracking, the sub-second movements of the animal in the odor landscape get superposed on the sniffing cycle. Given all this, it seems unlikely that the total number of spikes from a neuron in a 4-second period is going to be a relevant variable for neural processing downstream.

Much of the analysis focuses on the mean activity of the entire population. Why is this an interesting quantity? Apparently, the mean stays similar because some neurons increase and others decrease their firing rate. It would be more revealing, perhaps, to show the distribution of firing rates at different concentrations and see how that distribution is predicted by different models of normalization. This could provide a stronger test than just the mean.

The question "if concentration information is discarded in second-order neurons, which exclusively transmit odor information to the rest of the brain, how does the brain support olfactory behaviors, such as tracking and navigation?" is really not an open question anymore. For example, reference 23 reports in the abstract that "Odorant concentration had no systematic effect on spike counts, indicating that rate cannot encode intensity. Instead, odor intensity can be encoded by temporal features of the population response. We found a subpopulation of rapid, largely concentration-invariant responses was followed by another population of responses whose latencies systematically decreased at higher concentrations."

(3) Methods

It would be useful to state early in the manuscript what kinds of stimuli are being considered and how the response of a neuron is summarized by one number. There are many alternative ways to treat both stimuli and responses.

"The change in response across consecutive concentration levels may not be robust due to experimental noise and the somewhat limited range of concentrations sampled": Yes, a number of the curves just look like "no response". It would help the reader to show some examples of raw data, e.g. the time course of one neuron's firing rate to 4 concentrations, and for the authors to illustrate how they compress those responses into single numbers.

"We then calculated the angle between these two slopes for each neuron and plotted a polar histogram of these angles." The methods suggest that this angle is the arctan of the ratio of the two slopes in the response curve. A ratio of 2 would result from a slope change from 0.0001 to 0.0002 (i.e., virtually no change in slope) or from 1 to 2 (a huge change). Those are completely different response curves. Is it reasonable to lump them into the same bin of the polar plot? This seems an unusual way to illustrate the diversity of response curve shapes.

The *Drosophila* OSN data are passed through normalization models and then compared to locust PN data. This seems dangerous, as flies and locusts are separated by about 300 M years of evolution, and we don't know that fly PNs act like locust PNs. Their antennal lobe anatomy differs in many ways, as does the olfactory physiology. To draw any conclusions about a change in neural representation, it would be preferable to have OSN and PN data from the same species.

(4) Models of normalization

One conclusion is that divisive normalization could account for some of the change in responses from receptors to 2nd order neurons. This seems to be well appreciated already [e.g., Olsen 2010, Papadopoulou 2011, minireview in Hong & Wilson 2013].

Another claim is that subtractive normalization cannot perform that function. What model was used for subtractive normalization is unclear (there is an error in the Methods). It would be interesting if there were a categorical difference between divisive and subtractive normalization.

Looking closer at the divisive normalization model, it really has two components: (a) the "lateral inhibition" by which a neuron gets suppressed if other neurons fire (here scaled by the parameter k), and (b) a nonlinear sigmoid transformation (determined by the parameters n and σ). Both lateral inhibition and nonlinearity are known to contribute to decorrelation in a neural population (e.g., Pitkow 2012). The "intraglomerular gain control" contains only the nonlinearity. The "subtractive normalization" we don't know. But if one wanted to put divisive and subtractive inhibition on the same footing, one should add a sigmoid nonlinearity in both cases.

The response models could be made more realistic in other ways. For example, in both locusts and fish, the 2nd order neurons get inputs from multiple receptor types; presumably, that will affect their response functions. Also, lateral inhibition can take quite different forms. In locusts, the inhibitory neurons seem to collect from many glomeruli. But in rats, the inhibition by short axon cells may originate from just a few sparse glomeruli, and those might be different for every mitral cell (Fantana 2008).

(5) Tufted cells

There are questions raised by the following statements: "traded-off energy for faster and finer concentration discrimination" and "an additional type of second-order neuron (tufted cells) that has evolved in land vertebrates and that outperforms mitral cells in concentration encoding" and later "These results suggest a trade-off between concentration decoding and normalization processes, which prevent saturation and reduce energy consumption.". Are the tufted cells inferior to the mitral cells in any respect? Do they suffer from saturation at high concentration? And do they then fail in their postulated role for odor tracking? If not, then what was the evolutionary driver for normalization in the mitral cell pathway? Certainly not lower energy consumption (50,000 mitral cells = 1% of rod photoreceptors, each of which consumes way more energy than a mitral cell).

<https://doi.org/10.7554/eLife.107228.1.sa3>

Reviewer #2 (Public review):

Summary:

The main goal of this study is to examine how information about odor concentration is encoded by second-order neurons in the invertebrate and vertebrate olfactory system. In many animal models, the overall mean firing rates across the second-order neurons appear to be relatively flat or near constant with increasing odor intensity. While such compression of concentration information could aid in achieving concentration invariant recognition of odor identity, how this observation could be reconciled with the need to preserve information about the changes in stimulus intensity is a major focus of the study. The authors show that second-order neurons have 'diverse' dose-response curves and that the combinations of neurons activated (particularly the rank-order) differ with concentration. Further, they argue that a single circuit-level computation, termed 'divisive normalization,' where the individual neural response is normalized by the total activity across all neurons,

could help explain the coding properties of neurons at this stage of processing in all model organisms examined. They present approaches to read out the concentration information using spike rates or timing-based approaches. Finally, the authors reveal that tufted cells in the mouse olfactory bulb provide an exception to this coding approach and encode concentration information with a monotonic increase in firing rates.

Strengths:

- (1) Comparative analysis of odor intensity coding across four different species, revealing the common features in encoding stimulus-driven features, is highly valuable.
- (2) Showing how mitral and tufted cells differ in encoding odor intensity is potentially very important to the field.
- (3) How to preserve concentration information while compressing the same with divisive normalization is also a novel and important problem in the field of sensory coding.

Weaknesses:

(1) The encoding problem:

The main premise that divisive normalization generates this diversity of dose-response curves in the second-order neurons is a little problematic. The authors acknowledge this as part of their analysis in Figure 3.

"Therefore, divisive normalization mostly does not alter the relative contribution (rank order) of each neuron in the ensemble." (Page 4, last paragraph, lines 6-8).

The analysis in this figure indicates that divisive normalization does what it is supposed to do, i.e., compresses concentration information and not alter the rank-order of neurons or the combinatorial patterns. Changes in the combinations of neurons activated with intensity arise directly from the fact that the first-order neurons did not have monotonic responses with odor intensity (i.e., crossovers). This was the necessary condition, and not the divisive normalization for changes in the combinatorial code.

There seems to be a confusion/urge to attribute all coding properties found in the second-order neurons to 'divisive normalization.' If the input from sensory neurons is monotonic (i.e., no crossovers), then divisive normalization did not change the rank order, and the same combinations of neurons are activated in a similar fashion (same vector direction or combinatorial profile) to encode for different odor intensities. Concentration invariance is achieved, and concentration information is lost. However, when the first-order neurons are non-monotonic (i.e., with crossovers), that causes the second-order neurons to have different rank orders with different concentrations. Divisive normalization compresses information about concentrations, and rank-order differences preserve information about the odor concentration. Does this not mean that the non-monotonicity of sensory neuron response is vital for robustly maintaining information about odor concentration?

Naturally, the question that arises is whether many of the important features of the second-order neuron's response simply seem to follow the input. Or is my understanding of the figures and the write-up flawed, and are there more ways in which divisive normalization contributes to reshaping the second-order neural response? This must be clarified.

Lastly, the tufted cells in the mouse OB are also driven by this sensory input with crossovers. How does the OB circuit convert the input with crossovers into one that is monotonic with concentration? I think that is an important question that this computational effort could clarify.

(2) The decoding problem.

The way the decoding results and analysis are presented does not add a lot of information to what has already been presented. For example, based on the differences in rank-order with concentration, I would expect the combinatorial code to be different. Hence, a very simple classifier based on cosine or correlation distance would work well. However, since divisive normalization (DN) is applied, I would expect a simple classification scheme that uses the Euclidean distance metric to work equally as well after DN. Is this the case?

Leave-one-trial/sample-out seems too conservative. How robust are the combinatorial patterns across trials? Would just one or two training trials suffice for creating templates for robust classification? Based on my prior experience (<https://elifesciences.org/reviewed-preprints/89330>), I do expect that the combinatorial patterns would be more robust to adaptation and hence also allow robust recognition of odor intensity across repeated encounters.

Lastly, in the simulated data, since the affinity of the first-order sensory neurons to odorants is expected to be constant across concentration, and "Jaccard similarity between the sets of highest-affinity neurons for each pair of concentration levels was > 0.96 ," why would the rank-order change across concentration? DN should not alter the rank order.

If the set of early responders does change, how will the decoder need to change, and what precise predictions can be made that can be tested experimentally? The lack of exploration of this aspect of the results seems like a missed opportunity.

(3) Analysis of existing data.

I had a couple of issues related to the presentation and analysis of prior results.

i) Based on the methods, for Figures 1 and 2, it appears the responses across time, trials, and odorants were averaged to get a single data point per neuron for each concentration. Would this averaging not severely dilute trends in the data? The one that particularly concerns me is the averaging across different odorants. If you do odor-by-odor analysis, is the flattening of second-order neural responses still observable? Because some odorants activate more globally and some locally, I would expect a wide variety of dose-response relationships that vary with odor identity (more compressed in second-order neurons, of course). It would be good to show some representative neural responses and show how the extracted values for each neuron are a faithful/good representation of its response variation across intensities.

ii) A lot of neurons seem to have responses that flat line closer to zero (both firing rate and dF/F in Figure 1). Are these responsive neurons? The mean dF/F also seems to hover not significantly above zero. Hence, I was wondering if the number of neurons is reducing the trend in the data significantly.

iii) I did not fully understand the need to show the increase in the odor response across concentrations as a polar plot. I see potential issues with the same. For example, the following dose-response trend at four intensities (C4 being the highest concentration and C1 the lowest): response at C3 $>$ response at C1 and response at C4 $>$ response at C2. But response at C3 $<$ response at C2. Hence, it will be in the top right segment of the polar plot. However, the responses are not monotonic with concentrations. So, I am not convinced that the polar plot is the right way to characterize the dose-response curves. Just my 2 cents.

(4) Simulated vs. Actual data.

In many analyses, simulated data were used (Figures 3 and 4). However, there is no comparison of how well the simulated data fit the experimental data. For example, the Simulated 1st order neuron in Figure 3D does not show a change in rank-order for the first-order neuron. In Figure 3E, temporal response patterns in second-order neurons look

unrealistic. Some objective comparison of simulated and experimental data would help bolster confidence in these results.

<https://doi.org/10.7554/eLife.107228.1.sa2>

Reviewer #3 (Public review):

Summary:

In their study, Shen et al. examine how first- and second-order neurons of early olfactory circuits among invertebrates and vertebrates alike respond to and encode odor identity and concentration. Previously published electrophysiological and imaging data are re-analyzed and complemented with computational simulations. The authors explore multiple potential circuit computations by which odor concentration-dependent increases in first-order neuron responses transform into concentration-invariant responses on average across the second-order neuron population, and report that divisive normalization exceeds subtractive normalization and intraglomerular gain control in accounting for this transformation. The authors then explore how either rate- or timing-based schemes in third-order neurons may decode odor identity and concentration information from such concentration-invariant mean responses across the second-order neuron population. Finally, the results of their study of second-order neurons (invertebrate projection neurons and vertebrate mitral cells) are contrasted with the concentration-variant responses of second-order projection tufted cells in mammals. Overall, through a combination of neural data re-analysis, computational simulation, and conceptual theory, this study provides important new understanding of how aspects of sensory information are encoded through the actions of distinct components of early olfactory circuits.

Strengths:

Consideration of multiple evolutionarily disparate olfactory circuits, as well as re-analysis of previously published neural data sets combined with novel simulations guided by those sets, lends considerable robustness to some key findings of this study. In particular, the finding that divisive normalization - with direct inspiration from established circuit components in the form of glomerular layer short-axon cells - accounts more thoroughly for the average concentration invariance of second-order olfactory neurons at a population level than other forms of normalization is compelling. Likewise, demonstration of the required 'crossover' of first-order neuron concentration sensitivity for divisive normalization to achieve such flattening of concentration variance across the second-order population is notable, with simulations providing important insight into experimentally observed patterns of first-order neuron responses. Limited clarity in other aspects of the study, in particular related to the consideration of neural response latencies and enumerated below, temper the overall strength of the study.

Weaknesses:

(1) While the authors focus on concentration-dependent increases in first-order neuron activity, reflecting the majority of observed responses, recent work from the Imai group shows that odorants can also lead to direct first-order neuron inhibition (i.e., reduction in spontaneous activity), and within this subset, increasing odorant concentration tends to increase the degree of inhibition. Some discussion of these findings and how they may complement divisive normalization to contribute to the diverse second-order neuron concentration-dependence would be of interest and help expand the context of the current results.

(2) Related to the above point, odorant-evoked inhibition of second-order neurons is widespread in mammalian mitral cells and significantly contributes to the flattened concentration-dependence of mitral cells at the population level. Such responses are clearly seen in Figure 1D. Some discussion of how odorant-evoked mitral cell inhibition may complement divisive normalization, and likewise relate to comparatively lower levels of odorant-evoked inhibition among tufted cells, would further expand the context of the current results. Toward this end, replication of analyses in Figures 1D and E following exclusion of mitral cell inhibitory responses would provide insight into the contribution of such inhibition to the flattening of the mitral cell population concentration dependence.

(3) The idea of concentration-dependent crossover responses across the first-order population being required for divisive normalization to generate individually diverse concentration response functions across the second-order population is notable. The intuition of the crossover responses is that first-order neurons that respond most sensitively to any particular odorant (i.e., at the lowest concentration) respond with overall lower activity at higher concentrations than other first-order neurons less sensitively tuned to the odorant. Whether this is a consistent, generalizable property of odorant binding and first-order neuron responsiveness is not addressed by the authors, however. Biologically, one mechanism that may support such crossover events is intraglomerular presynaptic/feedback inhibition, which would be expected to increase with increasing first-order neuron activation such that the most-sensitively responding first-order neurons would also recruit the strongest inhibition as concentration increases, enabling other first-order neurons to begin to respond more strongly. Discussion of this and/or other biological mechanisms (e.g., first-order neuron depolarization block) supporting such crossover responses would strengthen these results.

(4) It is unclear to what degree the latency analysis considered in Figures 4D-H works with the overall framework of divisive normalization, which in Figure 3 we see depends on first-order neuron crossover in concentration response functions. Figure 4D suggests that all first-order neurons respond with the same response amplitude (R in eq. 3), even though this is supposed to be pulled from a distribution. It's possible that Figure 4D is plotting normalized response functions to highlight the difference in latency, but this is not clear from the plot or caption. If response amplitudes are all the same, and the response curves are, as plotted in Figure 4D, identical except for their time to half-max, then it seems somewhat trivial that the resulting second-order neuron activation will follow the same latency ranking, regardless of whether divisive normalization exists or not. However, there is some small jitter in these rankings across concentrations (Figure 4G), suggesting there is some randomness to the simulations. It would be helpful if this were clarified (e.g., by showing a non-normalized Figure 4D, with different response amplitudes), and more broadly, it would be extremely helpful in evaluating the latency coding within the broader framework proposed if the authors clarified whether the simulated first-order neuron response timecourses, when factoring in potentially different amplitudes (R) and averaging across the entire response window, reproduces the concentration response crossovers observed experimentally. In summary, in the present manuscript, it remains unclear if concentration crossovers are captured in the latency simulations, and if not, the authors do not clearly address what impact such variation in response amplitudes across concentrations may have on the latency results. It is further unclear to what degree divisive normalization is necessary for the second-order neurons to establish and maintain their latency ranks across concentrations, or to exhibit concentration-dependent changes in latency.

(5) How the authors get from Figure 4G to 4H is not clear. Figure 4G shows second-order neuron response latencies across all latencies, with ordering based on their sorted latency to low concentration. This shows that very few neurons appear to change latency ranks going from low to high concentration, with a change in rank appearing as any deviation in a monotonically increasing trend. Focusing on the high concentration points, there appear to

be 2 latency ranks switched in the first 10 responding neurons (reflecting the 1 downward dip in the points around neuron 8), rather than the 7 stated in the text. Across the first 50 responding neurons, I see only ~14 potential switches (reflecting the ~7 downward dips in the points around neurons 8, 20, 32, 33, 41, 44, 50), rather than the 32 stated in the text. It is possible that the unaccounted rank changes reflect fairly minute differences in latencies that are not visible in the plot in Figure 4G. This may be clarified by plotting each neuron's latency at low concentration vs. high concentration (i.e., similar to Figure 4H, but plotting absolute latency, not latency rank) to allow assessment of the absolute changes. If such minute differences are not driving latency rank changes in Fig. 4G, then a trend much closer to the unity line would be expected in Figure 4H. Instead, however, there are many massive deviations from unity, even within the first 50 responding neurons plotted in Figure 4G. These deviations include a jump in latency rank from 2 at low concentration to ~48 at high concentration. Such a jump is simply not seen in Figure 4G.

(6) In the text, the authors state that "Odor identity can be encoded by the set of highest-affinity neurons (which remains invariant across concentrations)." Presumably, this is a restatement of the primacy model and refers to invariance in latency rank (since the authors have not shown that the highest-affinity neurons have invariant response amplitudes across concentration). To what degree this statement holds given the results in Figure 4H, however, which appear to show that some neurons with the earliest latency rank at low concentration jump to much later latency ranks at high concentration, remains unclear. Such changes in latency rank for only a few of the first responding neurons may be negligible for classifying odor identity among a small handful of odorants, but not among 1-2 orders of magnitude more odors, which may feasibly occur in a natural setting. Collectively, these issues with the execution and presentation of the latency analysis make it unclear how robust the latency results are.

(7) Analysis in Figures 4A-C shows that concentration can be decoded from first-order neurons, second-order neurons, or first-order neurons with divisive normalization imposed (i.e., simulating second-order responses). This does not say that divisive normalization is necessary to encode concentration, however. Therefore, for the authors to say that divisive normalization is "a potential mechanism for generating odor-specific subsets of second-order neurons whose combinatorial activity or whose response latencies represent concentration information" seems too strong a conclusion. Divisive normalization is not generating the concentration information, since that can be decoded just as well from the first-order neurons. Rather, divisive normalization can account for the different population patterns in concentration response functions between first- and second-order neurons without discarding concentration-dependent information.

(8) Performing the same polar histogram analysis of tufted vs. mitral cell concentration response functions (Figure 5B) provides a compelling new visualization of how these two cell types differ in their concentration variance. The projected importance of tufted cells to navigation, emerging directly through the inverse relationship between average concentration and distance (Figure 5C), is not surprising, and is largely a conceptual analysis rather than new quantitative analysis per se, but nevertheless, this is an important point to make. Another important consideration absent from this section, however, is whether and how divisive normalization may impact tufted cell activity. Previous work from the authors, as well as from Schoppa, Shipley, and Westbrook labs, has compellingly demonstrated that a major circuit mediating divisive normalization of mitral cells (GABA/DAergic short-axon cells) directly targets external tufted cells, and is thus very likely to also influence projection tufted cells. Such analysis would additionally provide substantially more justification for the Discussion statement "we analyzed an additional type of second-order neuron (tufted cells)", which at present instead reflects fairly minimal analysis.

<https://doi.org/10.7554/eLife.107228.1.sa1>

Author response:

- (1) Explore the temporal component of neural responses (instead of collapsing responses to a single number, i.e., the average response over 4s), and determine which of the three models can recapitulate the observed dynamics.
- (2) Expand the polar plot visualization to show all three slopes (changes in responses across all three successive concentrations) instead of only two slopes.
- (3) Attempt to collect and analyze, from published papers, data of: (a) first-order neuron responses to odors to determine the role of first-order inhibition towards generating non-monotonic responses, and (b) PN responses in *Drosophila* to properly compare with corresponding first-order neuron responses.
- (4) Further discuss: (a) why the brain may need to encode absolute concentration, (b) the distinction between non-monotonic responses and cross-over responses, and (c) potential limitations of the primacy model.
- (5) Expand the divisive normalization model by evaluating different values of k and R , and study the effects of divisive normalization on tufted cells.
- (6) Add discussion of other potential inhibitory mechanisms that could contribute towards the observed effects.

Reviewer #1:

The article starts from the premise that animals need to know the absolute concentration of an odor over many log units, but the need for this isn't obvious. The introduction cites an analogy to vision and audition. These are cases where we know for a fact that the absolute intensity of the stimulus is not relevant. Instead, sensory perception relies on processing small differences in intensity across space or time. And to maintain that sensitivity to small differences, the system discards the stimulus baseline. Humans are notoriously bad at judging the absolute light level. That information gets discarded even before light reaches the retina, namely through contraction of the pupil. Similarly, it seems plausible that a behavior like olfactory tracking relies on sensing small gradients across time (when weaving back and forth across the track) or space (across nostrils). It is important that the system function over many log units of concentration (e.g., far and close to a source) but not that it accurately represents what that current concentration is [see e.g., Wachowiak et al, 2025 Recalibrating Olfactory Neuroscience..].

We thank the Reviewer for the insightful input and agree that gradients across time and space are important for various olfactory behaviors, such as tracking. At the same time, we think that absolute concentration is also needed for two reasons. First, in order to extract changes in concentration, the absolute concentration needs to be normalized out; i.e., change needs to be encoded with respect to some baseline, which is what divisive normalization computes. Second, while it is true that representing the exact number of odor molecules present is not important, this number directly relates to distance from the odor source, which does provide ethological value (e.g., is the tiger 100m or 1000m away?). Indeed, our decoding experiments focused on discriminating relative, and not on absolute, concentrations by classifying between each pair of concentrations (i.e., relative distances), which is effectively an assessment of the gradient. In our revision, we will make all of these points clearer.

Still, many experiments in olfactory research have delivered square pulses of odor at concentrations spanning many log units, rather than the sorts of stimuli an animal might encounter during tracking. Even within that framework, though, it doesn't seem mysterious anymore how odor identity and odor concentration are represented

differently. For example, Stopfer et al 2003 showed that the population response of locust PNs traces a dynamic trajectory. Trajectories for a given odor form a manifold, within which trajectories for different concentrations are distinct by their excursions on the manifold. To see this, one must recognize that the PN responds to an odor pulse with a time-varying firing rate, that different PNs have different dynamics, and that the dynamics can change with concentration. This is also well recognized in the mammalian systems. Much has been written about the topic of dynamic coding of identity and intensity - see the reviews of Laurent (2002) and Uchida (2014).

Given the above comments on the dynamics of odor responses in first- and second-order neurons, it seems insufficient to capture the response of a neuron with a single number. Even if one somehow had to use a single number, the mean firing rate during the odor pulse may not be the best choice. For example, the rodent mitral cells fire in rhythm with the animal's sniffing cycle, and certain odors will just shift the phase of the rhythm without changing the total number of spikes (see e.g., Fantana et al, 2008). During olfactory search or tracking, the sub-second movements of the animal in the odor landscape get superposed on the sniffing cycle. Given all this, it seems unlikely that the total number of spikes from a neuron in a 4-second period is going to be a relevant variable for neural processing downstream.

To our knowledge, it is not well understood how downstream brain regions read out mitral cell responses to guide olfactory behavior. The olfactory bulb projects to more than a dozen brain regions, and different regions could decode signals in different ways. We focused on the mean response because it is a simple, natural construct.

The datasets we analyzed may not include all relevant timing information; for example, the mouse data is from calcium imaging studies that did not track sniff timing. Nonetheless, we plan to address this comment within our framework by binning time into smaller-sized windows (e.g., 0-0.2s, 0.2-0.4s, etc.) and repeating our analysis for each of these windows. Specifically, we will determine how each normalization method fares in recapitulating statistics of the population responses of each window, beyond simply assessing the population mean.

Much of the analysis focuses on the mean activity of the entire population. Why is this an interesting quantity? Apparently, the mean stays similar because some neurons increase and others decrease their firing rate. It would be more revealing, perhaps, to show the distribution of firing rates at different concentrations and see how that distribution is predicted by different models of normalization. This could provide a stronger test than just the mean.

We agree that mean activity is only one measure to summarize a rich data set and will perform the suggested analysis.

The question "if concentration information is discarded in second-order neurons, which exclusively transmit odor information to the rest of the brain, how does the brain support olfactory behaviors, such as tracking and navigation?" is really not an open question anymore. For example, reference 23 reports in the abstract that "Odorant concentration had no systematic effect on spike counts, indicating that rate cannot encode intensity. Instead, odor intensity can be encoded by temporal features of the population response. We found a subpopulation of rapid, largely concentration-invariant responses was followed by another population of responses whose latencies systematically decreased at higher concentrations."

Primacy coding does provide one plausible mechanism to decode concentration. Our manuscript demonstrated how such a code could emerge in second-order neurons with the

help of divisive normalization, though it does require maintaining at least partial rank invariance across concentrations, which may not be robust. We also showed how concentration could be decoded via spike rates, even if average rates are constant, which provides an alternative hypothesis to that of ref 23.

Further, ref 23 only considers the piriform cortex, which, as mentioned above, is one of many targets of the olfactory bulb, and it remains unclear what the decoding mechanisms are of each of these targets. In addition, work from the same authors of ref 23 found multiple potential decoding strategies in the piriform cortex itself, including changes in firing rate (see Fig. 2E of ref. 23 - Bolding & Franks, 2017; as well as Fig. 4 in Roland et al., 2017).

It would be useful to state early in the manuscript what kinds of stimuli are being considered and how the response of a neuron is summarized by one number. There are many alternative ways to treat both stimuli and responses.

We will add this explanation to the manuscript.

"The change in response across consecutive concentration levels may not be robust due to experimental noise and the somewhat limited range of concentrations sampled": Yes, a number of the curves just look like "no response". It would help the reader to show some examples of raw data, e.g. the time course of one neuron's firing rate to 4 concentrations, and for the authors to illustrate how they compress those responses into single numbers.

We agree and will add this information to the manuscript.

"We then calculated the angle between these two slopes for each neuron and plotted a polar histogram of these angles." The methods suggest that this angle is the arctan of the ratio of the two slopes in the response curve. A ratio of 2 would result from a slope change from 0.0001 to 0.0002 (i.e., virtually no change in slope) or from 1 to 2 (a huge change). Those are completely different response curves. Is it reasonable to lump them into the same bin of the polar plot? This seems an unusual way to illustrate the diversity of response curve shapes.

We agree that the two changes in the reviewer's example will be categorized in the same quadrant in our analysis. We did not focus on the absolute changes because our analysis covers many log ratios of concentrations. Instead, we focused on the relative shapes of the concentration response curves, and more specifically, the direction of the change (i.e., the sign of the slope). We will better motivate this style of analysis in the revision. Moreover, in response to comments by Reviewer 2, we will compare response shapes between all three successive levels of concentration changes, as opposed to only two levels.

The Drosophila OSN data are passed through normalization models and then compared to locust PN data. This seems dangerous, as flies and locusts are separated by about 300 M years of evolution, and we don't know that fly PNs act like locust PNs. Their antennal lobe anatomy differs in many ways, as does the olfactory physiology. To draw any conclusions about a change in neural representation, it would be preferable to have OSN and PN data from the same species.

We are in the process of requesting PN response data in Drosophila from groups that have collected such data and will repeat the analysis once we get access to the data.

One conclusion is that divisive normalization could account for some of the change in responses from receptors to 2nd order neurons. This seems to be well appreciated

already [e.g., Olsen 2010, Papadopoulou 2011, minireview in Hong & Wilson 2013].

While we agree that these manuscripts do study the effects of divisive normalization in insects and fish, here we show that this computation also generalizes to rodents. In addition, these previous studies do not focus on divisive normalization's role towards concentration encoding/decoding, which is our focus. We will clarify this difference in the revision.

Another claim is that subtractive normalization cannot perform that function. What model was used for subtractive normalization is unclear (there is an error in the Methods). It would be interesting if there were a categorical difference between divisive and subtractive normalization.

We apologize for the mistake in the subtractive normalization equation and will correct it. Thank you for catching it.

Looking closer at the divisive normalization model, it really has two components: (a) the "lateral inhibition" by which a neuron gets suppressed if other neurons fire (here scaled by the parameter k), and (b) a nonlinear sigmoid transformation (determined by the parameters n and σ). Both lateral inhibition and nonlinearity are known to contribute to decorrelation in a neural population (e.g., Pitkow 2012). The "intraglomerular gain control" contains only the nonlinearity. The "subtractive normalization" we don't know. But if one wanted to put divisive and subtractive inhibition on the same footing, one should add a sigmoid nonlinearity in both cases.

Our intent was not to place all the methods on the "same footing" but rather to isolate the two primary components of normalization methods – non-linearity and lateral inhibition – and determine which of these, and in which combination, could generate the desired effects. Divisive normalization incorporates both components, whereas intraglomerular gain control and subtractive normalization only incorporate one of these components. We will clarify this reasoning in the revision.

The response models could be made more realistic in other ways. For example, in both locusts and fish, the 2nd order neurons get inputs from multiple receptor types; presumably, that will affect their response functions. Also, lateral inhibition can take quite different forms. In locusts, the inhibitory neurons seem to collect from many glomeruli. But in rats, the inhibition by short axon cells may originate from just a few sparse glomeruli, and those might be different for every mitral cell (Fantana 2008).

We thank the Reviewer for the input. Instead of fixing k for all second-order neurons, we will apply different k values for different neurons. We will also systematically vary the percentage of neurons used for the divisive normalization calculation in the denominator, and determine the regime under which the effects experimentally observed are reproducible. This approach takes into account the scenario that inter-glomerular inhibitory interactions are sparse.

There are questions raised by the following statements: "traded-off energy for faster and finer concentration discrimination" and "an additional type of second-order neuron (tufted cells) that has evolved in land vertebrates and that outperforms mitral cells in concentration encoding" and later "These results suggest a trade-off between concentration decoding and normalization processes, which prevent saturation and reduce energy consumption.". Are the tufted cells inferior to the mitral cells in any respect? Do they suffer from saturation at high concentration? And do they then fail in their postulated role for odor tracking? If not, then what was the evolutionary driver for normalization in the mitral cell pathway? Certainly not lower energy consumption

(50,000 mitral cells = 1% of rod photoreceptors, each of which consumes way more energy than a mitral cell).

The question of what mitral cells are “good for”, compared to tufted cells, remains unclear in our view. We speculate that mitral cells provide superior context-dependent processing and are better for determining stimuli-reward contingencies, but this remains far from settled experimentally.

We believe the mitral cell pathway evolved earlier than tufted cells, since the former appear akin to projection neurons in insects. Nonetheless, we agree that differences in energy consumption are unlikely to be the primary distinguishing factor, and in the revision, we will drop this argument.

Reviewer #2:

The main premise that divisive normalization generates this diversity of dose-response curves in the second-order neurons is a little problematic. ... The analysis in [Figure 3] indicates that divisive normalization does what it is supposed to do, i.e., compresses concentration information and not alter the rank-order of neurons or the combinatorial patterns. Changes in the combinations of neurons activated with intensity arise directly from the fact that the first-order neurons did not have monotonic responses with odor intensity (i.e., crossovers). This was the necessary condition, and not the divisive normalization for changes in the combinatorial code. There seems to be a confusion/urge to attribute all coding properties found in the second-order neurons to 'divisive normalization.' If the input from sensory neurons is monotonic (i.e., no crossovers), then divisive normalization did not change the rank order, and the same combinations of neurons are activated in a similar fashion (same vector direction or combinatorial profile) to encode for different odor intensities. Concentration invariance is achieved, and concentration information is lost. However, when the first-order neurons are non-monotonic (i.e., with crossovers), that causes the second-order neurons to have different rank orders with different concentrations. Divisive normalization compresses information about concentrations, and rank-order differences preserve information about the odor concentration. Does this not mean that the non-monotonicity of sensory neuron response is vital for robustly maintaining information about odor concentration? Naturally, the question that arises is whether many of the important features of the second-order neuron's response simply seem to follow the input. Or is my understanding of the figures and the write-up flawed, and are there more ways in which divisive normalization contributes to reshaping the second-order neural response? This must be clarified. Lastly, the tufted cells in the mouse OB are also driven by this sensory input with crossovers. How does the OB circuit convert the input with crossovers into one that is monotonic with concentration? I think that is an important question that this computational effort could clarify.

It appears that there is confusion about the definitions of “non-monotonicity” and “crossovers”. These are two independent concepts – one does not necessarily lead to the other. Non-monotonicity concerns the response of a single neuron to different concentration levels. A neuron’s response is considered non-monotonic if its response goes up then down, or down then up, across increasing concentrations. A “cross-over” is defined based on the responses of multiple neurons. A cross-over occurs when the response of one neuron is lower than another neuron at one concentration, but higher than the other at a different concentration. For example, the responses of both neurons could increase monotonically with increasing concentration, but one neuron might start lower and grow faster, hence creating a cross-over. We will clarify this in the manuscript, which we believe will resolve the questions raised above.

The way the decoding results and analysis are presented does not add a lot of information to what has already been presented. For example, based on the differences in rank-order with concentration, I would expect the combinatorial code to be different. Hence, a very simple classifier based on cosine or correlation distance would work well. However, since divisive normalization (DN) is applied, I would expect a simple classification scheme that uses the Euclidean distance metric to work equally as well after DN. Is this the case?

Yes, we used a simple classification scheme, logistic regression with a linear kernel, which is essentially a Euclidean distance-based classification. This scheme works better for tufted cells because they are more monotonic; i.e., if neuron A and B both increase their responsiveness with concentration, then Euclidean distance would be fine. But if neuron A's response amplitude goes up and neuron B's response goes down – as often happens for mitral cells – then Euclidean distance does not work as well. We will add intuition about this in the manuscript.

Leave-one-trial/sample-out seems too conservative. How robust are the combinatorial patterns across trials? Would just one or two training trials suffice for creating templates for robust classification? Based on my prior experience (<https://elifesciences.org/reviewed-preprints/89330>), I do expect that the combinatorial patterns would be more robust to adaptation and hence also allow robust recognition of odor intensity across repeated encounters.

As suggested, we will compute the correlation coefficient of the similarity of neural responses for each odor (across trials). We will repeat this analysis for both mitral and tufted cells. To determine the effect of adaptation, we will compute correlation coefficients of responses between the 1st and 2nd trials vs the 1st and final trial.

Lastly, in the simulated data, since the affinity of the first-order sensory neurons to odorants is expected to be constant across concentration, and "Jaccard similarity between the sets of highest-affinity neurons for each pair of concentration levels was > 0.96," why would the rank-order change across concentration? DN should not alter the rank order.

We agree that divisive normalization should not alter the rank order, but the rank order may change in first-order neurons, which carries through to second-order neurons. This confusion may be related to the one mentioned above re: cross-overs vs non-monotonicity. Moreover, in the simulated data (Fig. 4D-H), the Jaccard similarity was calculated based on only the 50 neurons with the highest affinity, not the entire population of neurons. As shown in Fig. 4H, most of the rank-order change happens in the remaining 150 neurons.

Note that in response to a comment by Reviewer 3, we will change the presentation of Fig. 4H in the revision.

If the set of early responders does change, how will the decoder need to change, and what precise predictions can be made that can be tested experimentally? The lack of exploration of this aspect of the results seems like a missed opportunity.

In the Discussion, we wrote about how downstream circuits will need to learn which set of neurons are to be associated with each distinct concentration level. We will expand upon this point and include experimentally testable predictions.

Based on the methods, for Figures 1 and 2, it appears the responses across time, trials, and odorants were averaged to get a single data point per neuron for each concentration. Would this averaging not severely dilute trends in the data? The one that particularly concerns me is the averaging across different odorants. If you do odor-by-odor analysis, is the flattening of second-order neural responses still observable? Because some odorants activate more globally and some locally, I would expect a wide variety of dose-response relationships that vary with odor identity (more compressed in second-order neurons, of course). It would be good to show some representative neural responses and show how the extracted values for each neuron are a faithful/good representation of its response variation across intensities.

It appears there is some confusion here; we will clarify in the text and figure captions that we did not average across different odors in our analysis. We will also add figure panels showing some representative neural responses as suggested by the Reviewer.

A lot of neurons seem to have responses that flat line closer to zero (both firing rate and dF/F in Figure 1). Are these responsive neurons? The mean dF/F also seems to hover not significantly above zero. Hence, I was wondering if the number of neurons is reducing the trend in the data significantly.

Yes, if a neuron responds to at least one concentration level in at least 50% of the trials, it is considered responsive. So it is possible that some neurons respond to one concentration level and otherwise flatline near zero. We will highlight a few example neurons to visualize this scenario.

I did not fully understand the need to show the increase in the odor response across concentrations as a polar plot. I see potential issues with the same. For example, the following dose-response trend at four intensities (C4 being the highest concentration and C1 the lowest): response at C3 > response at C1 and response at C4 > response at C2. But response at C3 < response at C2. Hence, it will be in the top right segment of the polar plot. However, the responses are not monotonic with concentrations. So, I am not convinced that the polar plot is the right way to characterize the dose-response curves. Just my 2 cents.

Your 2 cents are valuable! Thank you for raising this point. Instead of computing two slopes (C1-C3 and C2-C4), we will expand our analysis to include all three slopes (C1-C2, C2-C3, C3-C4). Consequently, there are $2^3 = 8$ different response shapes, and we will list them and quantify the fraction of the responses that fall into each shape category.

In many analyses, simulated data were used (Figures 3 and 4). However, there is no comparison of how well the simulated data fit the experimental data. For example, the Simulated 1st order neuron in Figure 3D does not show a change in rank-order for the first-order neuron. In Figure 3E, temporal response patterns in second-order neurons look unrealistic. Some objective comparison of simulated and experimental data would help bolster confidence in these results.

We believe the Reviewer is referring to Figs. 4D and 4E, since Fig. 3D does not show a first-order neuron simulation, and there is no Fig 3E. In Fig. 4D there is no change of rank order because the simulation is for a single odor and single concentration level, and the change of rank-order (i.e., cross-overs) as we define occurs between concentration levels. We will clarify this in the manuscript.

Reviewer #3:

While the authors focus on concentration-dependent increases in first-order neuron activity, reflecting the majority of observed responses, recent work from the Imai group shows that odorants can also lead to direct first-order neuron inhibition (i.e., reduction in spontaneous activity), and within this subset, increasing odorant concentration tends to increase the degree of inhibition. Some discussion of these findings and how they may complement divisive normalization to contribute to the diverse second-order neuron concentration-dependence would be of interest and help expand the context of the current results.

We thank the Reviewer for the suggestion. We will request datasets of first-order neuron responses from the groups who acquired them. We will analyze this data to determine the role of inhibition or antagonistic binding and quantify what percentage of first-order neurons respond less strongly with larger concentrations.

Related to the above point, odorant-evoked inhibition of second-order neurons is widespread in mammalian mitral cells and significantly contributes to the flattened concentration-dependence of mitral cells at the population level. Such responses are clearly seen in Figure 1D. Some discussion of how odorant-evoked mitral cell inhibition may complement divisive normalization, and likewise relate to comparatively lower levels of odorant-evoked inhibition among tufted cells, would further expand the context of the current results. Toward this end, replication of analyses in Figures 1D and E following exclusion of mitral cell inhibitory responses would provide insight into the contribution of such inhibition to the flattening of the mitral cell population concentration dependence.

We will perform the analysis suggested, specifically, we will set the negative mitral cell responses to 0 and assess whether the population mean remains flat.

The idea of concentration-dependent crossover responses across the first-order population being required for divisive normalization to generate individually diverse concentration response functions across the second-order population is notable. The intuition of the crossover responses is that first-order neurons that respond most sensitively to any particular odorant (i.e., at the lowest concentration) respond with overall lower activity at higher concentrations than other first-order neurons less sensitively tuned to the odorant. Whether this is a consistent, generalizable property of odorant binding and first-order neuron responsiveness is not addressed by the authors, however. Biologically, one mechanism that may support such crossover events is intraglomerular presynaptic/feedback inhibition, which would be expected to increase with increasing first-order neuron activation such that the most-sensitively responding first-order neurons would also recruit the strongest inhibition as concentration increases, enabling other first-order neurons to begin to respond more strongly. Discussion of this and/or other biological mechanisms (e.g., first-order neuron depolarization block) supporting such crossover responses would strengthen these results.

We thank the reviewer for providing additional mechanisms to consider. As suggested, we will add discussion of these alternatives to divisive normalization.

It is unclear to what degree the latency analysis considered in Figures 4D-H works with the overall framework of divisive normalization, which in Figure 3 we see depends on first-order neuron crossover in concentration response functions. Figure 4D suggests that all first-order neurons respond with the same response amplitude (R in eq. 3), even though this is supposed to be pulled from a distribution. It's possible that Figure 4D is

plotting normalized response functions to highlight the difference in latency, but this is not clear from the plot or caption. If response amplitudes are all the same, and the response curves are, as plotted in Figure 4D, identical except for their time to half-max, then it seems somewhat trivial that the resulting second-order neuron activation will follow the same latency ranking, regardless of whether divisive normalization exists or not. However, there is some small jitter in these rankings across concentrations (Figure 4G), suggesting there is some randomness to the simulations. It would be helpful if this were clarified (e.g., by showing a non-normalized Figure 4D, with different response amplitudes), and more broadly, it would be extremely helpful in evaluating the latency coding within the broader framework proposed if the authors clarified whether the simulated first-order neuron response timecourses, when factoring in potentially different amplitudes (R) and averaging across the entire response window, reproduces the concentration response crossovers observed experimentally. In summary, in the present manuscript, it remains unclear if concentration crossovers are captured in the latency simulations, and if not, the authors do not clearly address what impact such variation in response amplitudes across concentrations may have on the latency results. It is further unclear to what degree divisive normalization is necessary for the second-order neurons to establish and maintain their latency ranks across concentrations, or to exhibit concentration-dependent changes in latency.

As suggested by the Reviewer, we will add another simulation scenario where the response amplitudes (R) are different for different neurons. For each concentration, we will then average each neuron's response across the entire response window and determine if the simulation reproduces the cross-overs as observed experimentally.

How the authors get from Figure 4G to 4H is not clear. Figure 4G shows second-order neuron response latencies across all latencies, with ordering based on their sorted latency to low concentration. This shows that very few neurons appear to change latency ranks going from low to high concentration, with a change in rank appearing as any deviation in a monotonically increasing trend. Focusing on the high concentration points, there appear to be 2 latency ranks switched in the first 10 responding neurons (reflecting the 1 downward dip in the points around neuron 8), rather than the 7 stated in the text. Across the first 50 responding neurons, I see only ~14 potential switches (reflecting the ~7 downward dips in the points around neurons 8, 20, 32, 33, 41, 44, 50), rather than the 32 stated in the text. It is possible that the unaccounted rank changes reflect fairly minute differences in latencies that are not visible in the plot in Figure 4G. This may be clarified by plotting each neuron's latency at low concentration vs. high concentration (i.e., similar to Figure 4H, but plotting absolute latency, not latency rank) to allow assessment of the absolute changes. If such minute differences are not driving latency rank changes in Fig. 4G, then a trend much closer to the unity line would be expected in Figure 4H. Instead, however, there are many massive deviations from unity, even within the first 50 responding neurons plotted in Figure 4G. These deviations include a jump in latency rank from 2 at low concentration to ~48 at high concentration. Such a jump is simply not seen in Figure 4G.

We apologize that Fig. 4H was a poor choice for visualization. What is plotted in Fig. 4H is the sorted identity of neurons under low and high concentrations, and points on the $y=x$ line indicate that the two corresponding neurons have the same rank under the two concentrations. We will replace this panel with a more intuitive visualization, where the x and y axes are the ranks of the neurons; and deviation from the $y=x$ line indicates how different the ranks are of a neuron to the two concentrations.

In the text, the authors state that "Odor identity can be encoded by the set of highest-affinity neurons (which remains invariant across concentrations)." Presumably, this is a

restatement of the primacy model and refers to invariance in latency rank (since the authors have not shown that the highest-affinity neurons have invariant response amplitudes across concentration). To what degree this statement holds given the results in Figure 4H, however, which appear to show that some neurons with the earliest latency rank at low concentration jump to much later latency ranks at high concentration, remains unclear. Such changes in latency rank for only a few of the first responding neurons may be negligible for classifying odor identity among a small handful of odorants, but not among 1-2 orders of magnitude more odors, which may feasibly occur in a natural setting. Collectively, these issues with the execution and presentation of the latency analysis make it unclear how robust the latency results are.

The original primacy model states that the latency of a neuron decreases with increasing concentration, while the ranks of neurons remain unaltered. Our results, on the other hand, suggest that the ranks do at least partially change across concentrations. This leads to two possible decoding mechanisms. First, if the top K responding neurons remain invariant across concentrations (even if their individual ranks change within the top K), then the brain could learn to associate a population of K neurons with a response latency; lower response latency means higher concentration. Second, if the top K responding neurons do not remain invariant across concentrations, then the brain would need to learn to associate a different set of neurons with each concentration level. The latter imposes additional constraints on the robustness of the primacy model and the corresponding read-out mechanism. We will include more discussion of these possibilities in the revision.

Analysis in Figures 4A-C shows that concentration can be decoded from first-order neurons, second-order neurons, or first-order neurons with divisive normalization imposed (i.e., simulating second-order responses). This does not say that divisive normalization is necessary to encode concentration, however. Therefore, for the authors to say that divisive normalization is "a potential mechanism for generating odor-specific subsets of second-order neurons whose combinatorial activity or whose response latencies represent concentration information" seems too strong a conclusion. Divisive normalization is not generating the concentration information, since that can be decoded just as well from the first-order neurons. Rather, divisive normalization can account for the different population patterns in concentration response functions between first- and second-order neurons without discarding concentration-dependent information.

We agree that the word "generating" is faulty. We thank the reviewer for their more precise wording, which we will adopt.

Performing the same polar histogram analysis of tufted vs. mitral cell concentration response functions (Figure 5B) provides a compelling new visualization of how these two cell types differ in their concentration variance. The projected importance of tufted cells to navigation, emerging directly through the inverse relationship between average concentration and distance (Figure 5C), is not surprising, and is largely a conceptual analysis rather than new quantitative analysis per se, but nevertheless, this is an important point to make. Another important consideration absent from this section, however, is whether and how divisive normalization may impact tufted cell activity. Previous work from the authors, as well as from Schoppa, Shipley, and Westbrook labs, has compellingly demonstrated that a major circuit mediating divisive normalization of mitral cells (GABA/DAergic short-axon cells) directly targets external tufted cells, and is thus very likely to also influence projection tufted cells. Such analysis would additionally provide substantially more justification for the Discussion statement "we analyzed an additional type of second-order neuron (tufted cells)", which at present instead reflects fairly minimal analysis.

We agree that tufted cells are subject to divisive normalization as well, albeit probably to a less degree than mitral cells. To determine the effect of this, we will alter the strength (and degree of sparseness of interglomerular interactions) of divisive normalization and determine if there is a regime where response features of tufted cells match those observed experimentally.

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